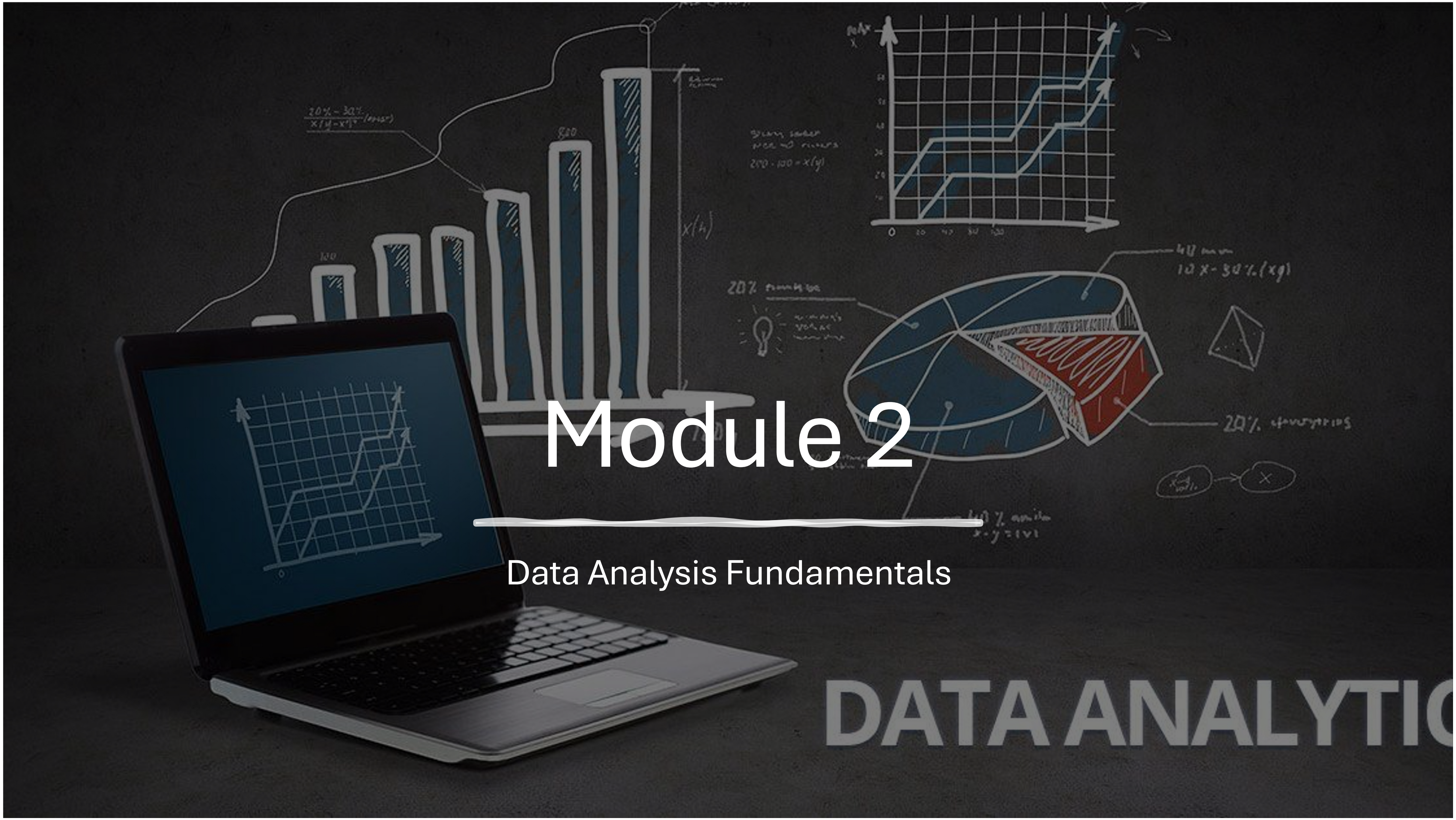
K-Youth Training Program

Data Analytics for business growth and sustainability in the workforce

Module 2

Data Analysis Fundamentals



slido

Please download and install the
Slido app on all computers you use

What did you find interesting about yesterday's class?

ⓘ Start presenting to display the poll results on this slide.

Module Contents

Module 2: Data Analysis Fundamentals

- 1. Databases – types and purpose
- 2. Tools used to ingest data
- 3. Storing and retrieving data
- 4. Data Quality and Cleaning Techniques
- 5. Data Transformation
- 6. Data Integration and ETL Processes
- 7. Python Basics for Non-IT Students
- a. Introduction to Python and Its Uses in Data Analytics
- b. Setting Up Python Environment
- c. Basic Python Syntax and Operations
- d. Data Structures: Lists, Tuples, Dictionaries
- e. Importing and Using Libraries (NumPy, Pandas)
- f. Basic Data Manipulation with Pandas

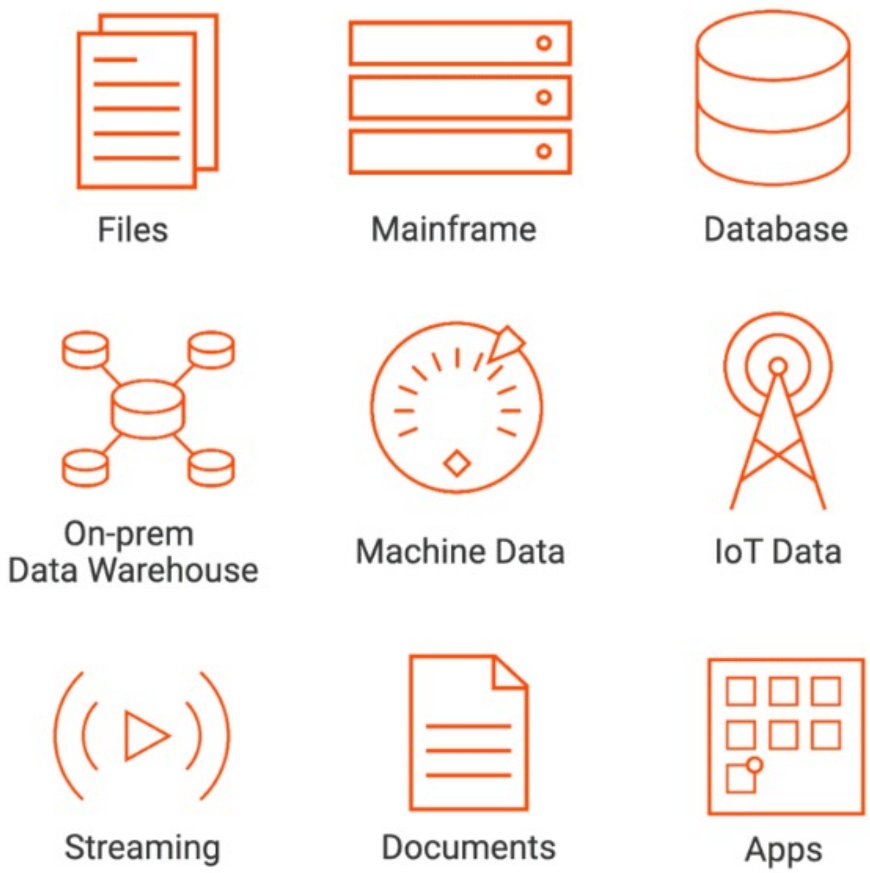
Data Integration and ETL

Data integration and ETL (Extract, Transform, Load) are essential processes in the data analytics pipeline. They ensure that data from disparate sources is combined, cleaned, and structured in a way that makes it usable for analysis, reporting, and decision-making.

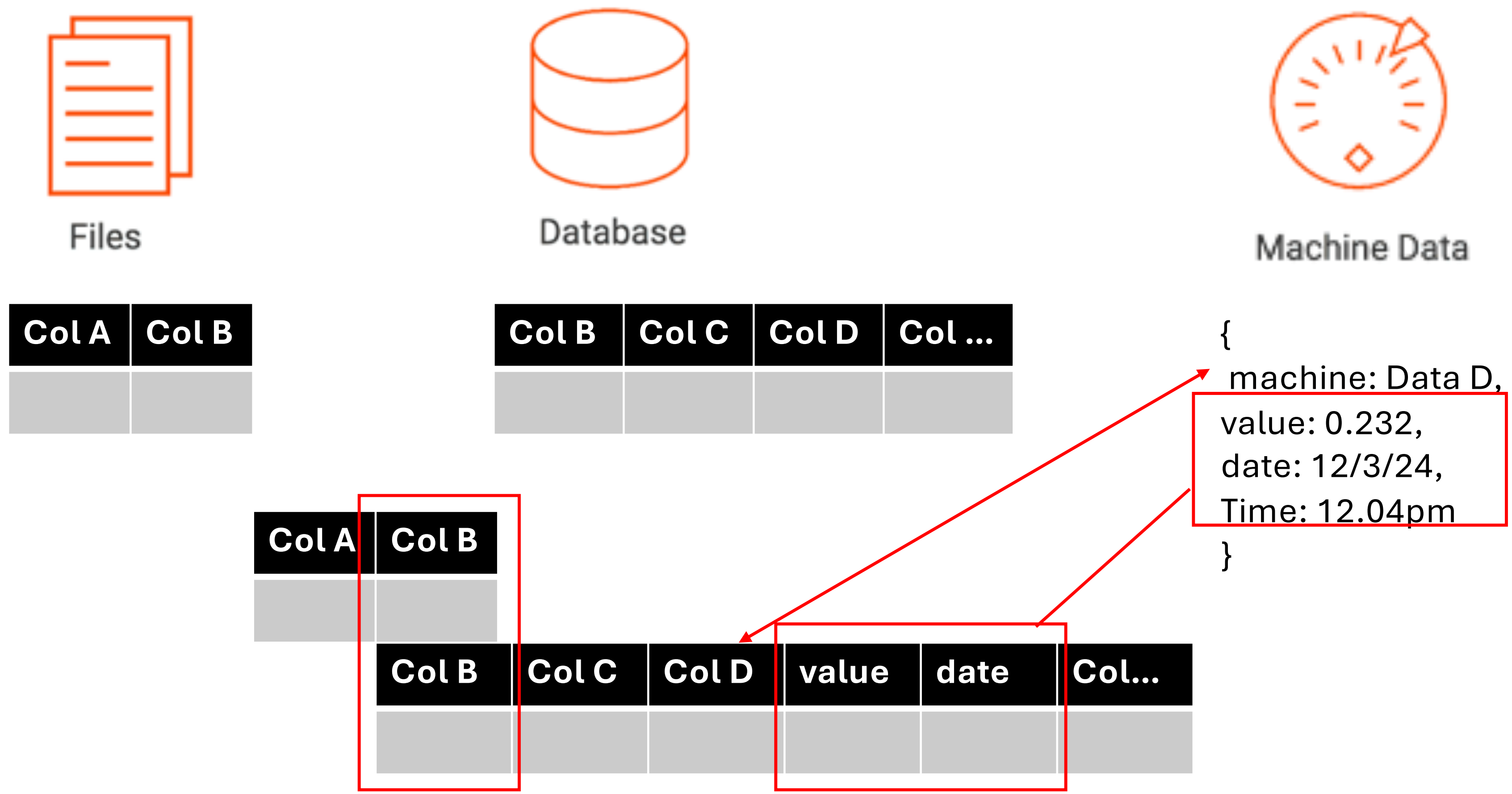
Key steps / activities include the following:

1. **Data Source Identification**
2. **Data Mapping**
3. **Data Fusion**
4. **Schema Reconciliation**
5. **Conflict Resolution**

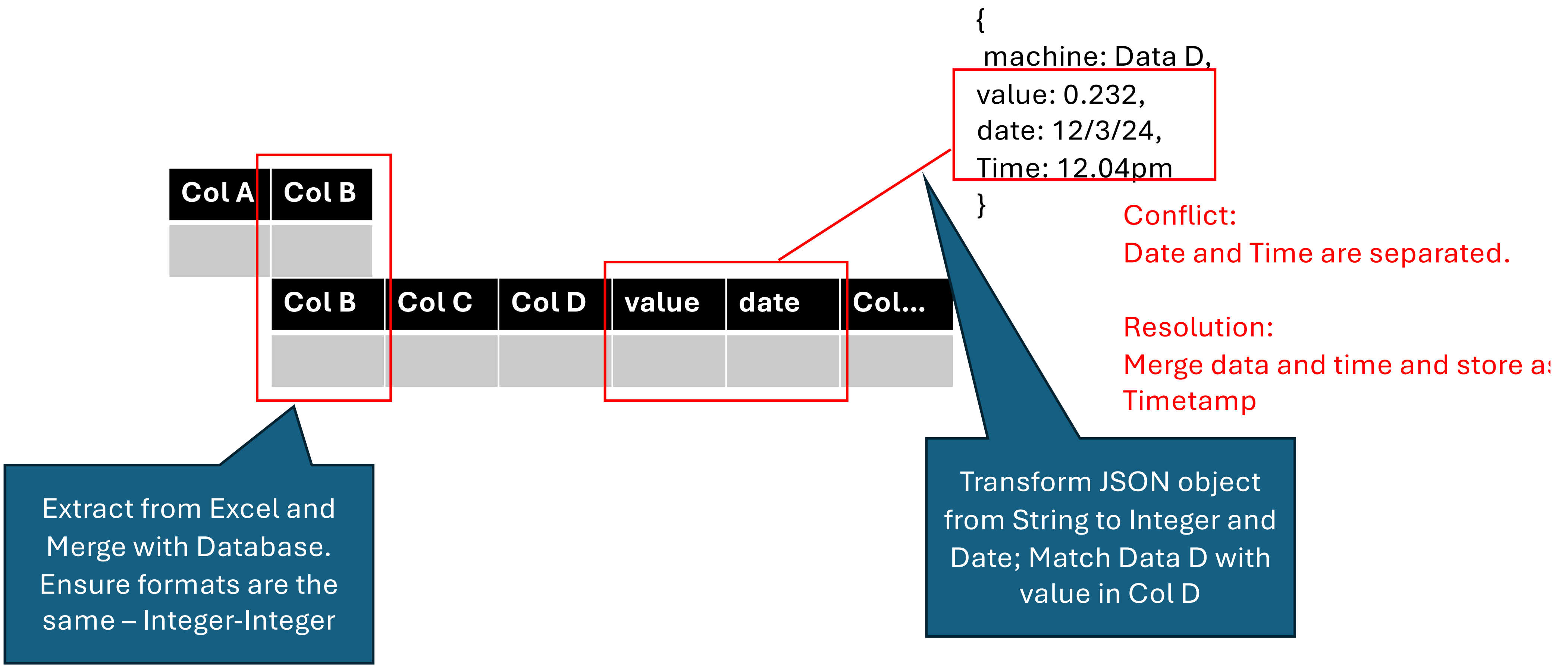
Key steps – Data Source Identification



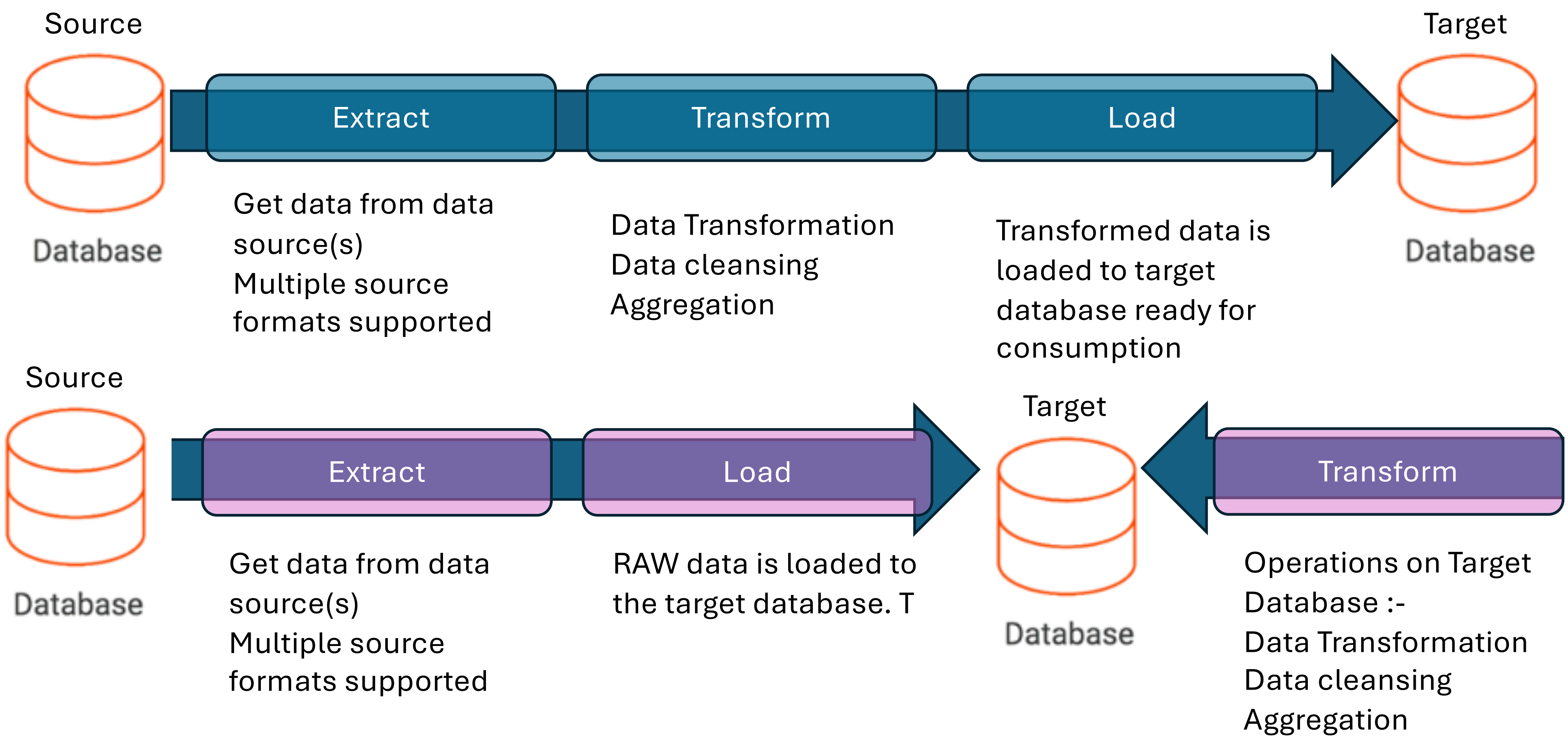
Key steps – Data Mapping and Fusion



Key steps – Schema Reconciliation



Extract Transform Load (ELT) process



Comparison between ETL and ELT

Aspect	ETL	ELT
Transformation Timing	Before loading into the target	After loading into the target
Target System	Traditional databases, data warehouses	Cloud-based systems (e.g., Snowlake, BigQuery)
Scalability	Limited scalability with large data	Highly scalable for big data
Processing Location	Intermediate ETL tools or servers	Target system (leverages cloud power)
Data Load Type	Processed data is loaded	Raw data is loaded
Flexibility	Less flexible for schema changes	More flexible and agile for transformations
Performance	Slower with large datasets	Optimized for large datasets
Cost	May require additional ETL tools	Reduced tool dependency; higher cloud storage costs